JAVA ARRAY ASSIGNMENT

Q2.

CODE:

```
import java.util.Scanner;
public class array1 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter size of array: ");
        int n = sc.nextInt();
        int arr[] = new int[n];
        System.out.println("Enter array elements:");
        for(int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }
        for(int i = 0; i < n; i++) {
            for(int j = i + 1; j < n; j++) {
                if(arr[i] > arr[j]) {
                    int temp = arr[i];
                    arr[i] = arr[j];
                    arr[j] = temp;
                }
            }
        }
        System.out.println("Sorted array:");
        for(int i = 0; i < n; i++) {
            System.out.print(arr[i] + " ");
        }
        sc.close();
    }
}
```

OUTPUT:

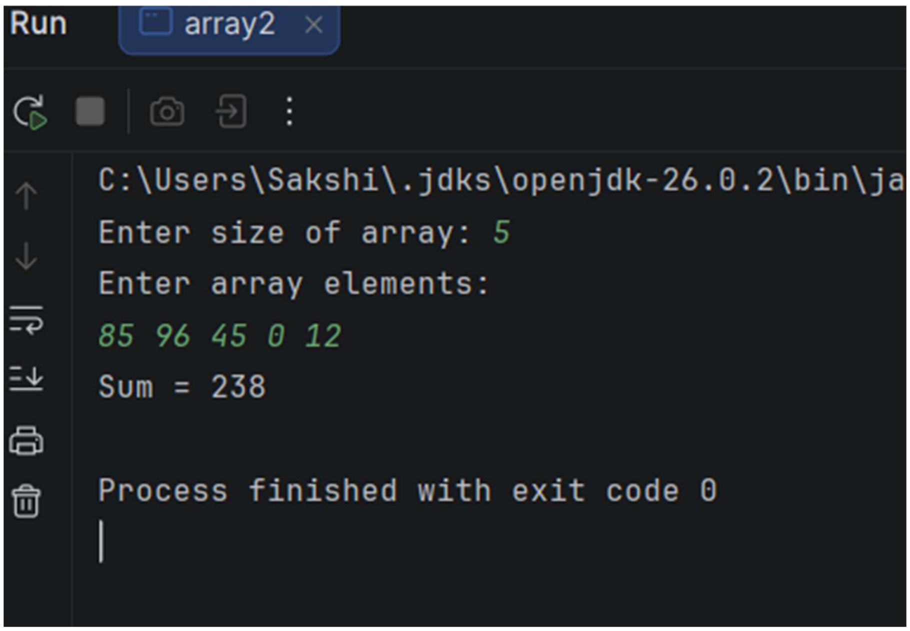
Q3.

CODE:

```
import java.util.Scanner;
public class array2 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter size of array: ");
        int n = sc.nextInt();
        int arr[] = new int[n];
        System.out.println("Enter array elements:");
        for(int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }
        int sum = 0;
        for(int i = 0; i < n; i++) {
```

```
            sum = sum + arr[i];
        }
        System.out.println("Sum = " + sum);
        sc.close();
    }
}
```

OUTPUT:



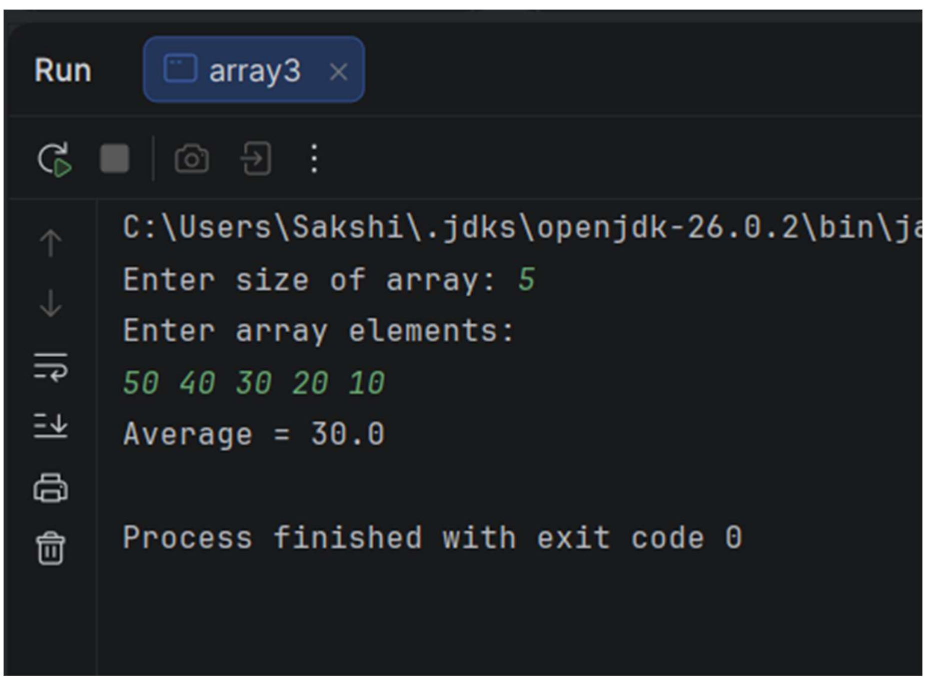
Q4.

CODE:

```
import java.util.Scanner;
public class array3 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter size of array: ");
        int n = sc.nextInt();
        int arr[] = new int[n];
```

```
        System.out.println("Enter array elements:");
        for(int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }
        int sum = 0;
        for(int i = 0; i < n; i++) {
            sum = sum + arr[i];
        }
        double average = (double) sum / n;
        System.out.println("Average = " + average);
        sc.close();
    }
}
```

OUTPUT:

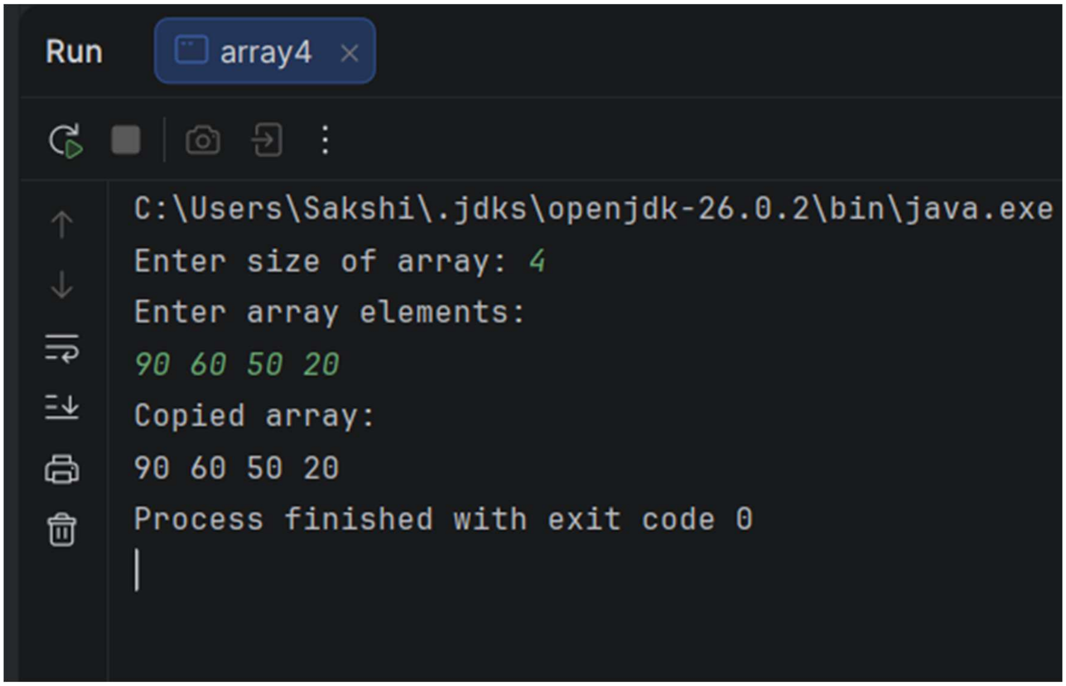


Q5.

CODE:

```java
import java.util.Scanner;
public class array4 {
	public static void main(String[] args) {
	    Scanner sc = new Scanner(System.in);
	    System.out.print("Enter size of array: ");
	    int n = sc.nextInt();
	    int arr[] = new int[n];
	    int copy[] = new int[n];
	    System.out.println("Enter array elements:");
	    for(int i = 0; i < n; i++) {
	        arr[i] = sc.nextInt();
	    }
	    for(int i = 0; i < n; i++) {
	        copy[i] = arr[i];
	    }
	    System.out.println("Copied array:");
	    for(int i = 0; i < n; i++) {
	        System.out.print(copy[i] + " ");
	    }
	    sc.close();
	}
  }
```

OUTPUT:



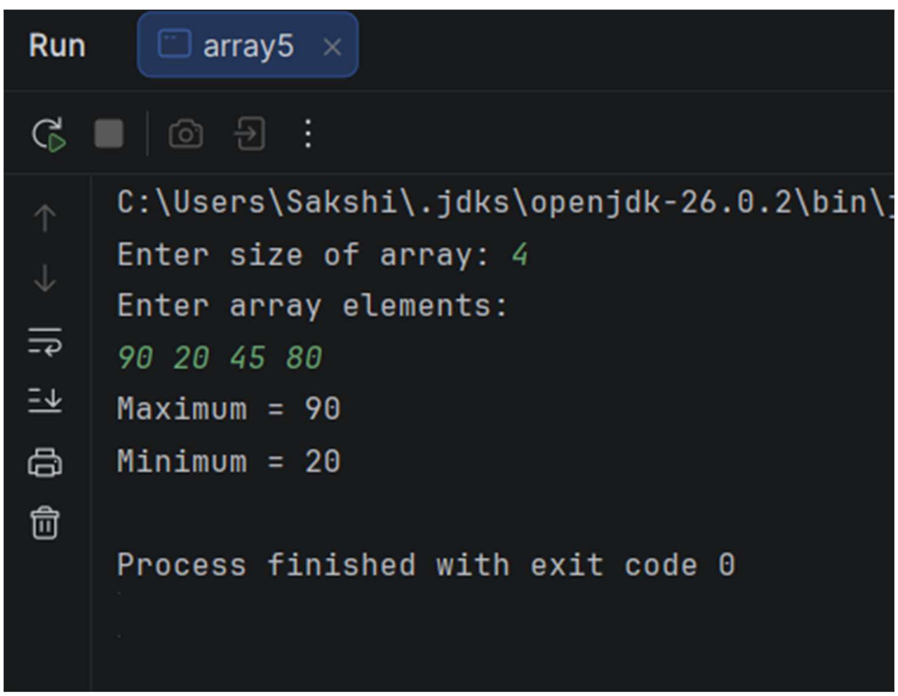
Q6.

CODE:

```
import java.util.Scanner;
public class array5 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter size of array: ");
        int n = sc.nextInt();
        int arr[] = new int[n];
        System.out.println("Enter array elements:");
        for(int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }
        int max = arr[0];
        int min = arr[0];
        for(int i = 1; i < n; i++) {
            if(arr[i] > max) {
```

```
            max = arr[i];
        }
        if(arr[i] < min) {
            min = arr[i];
        }
    }
    System.out.println("Maximum = " + max);
    System.out.println("Minimum = " + min);
    sc.close();
  }
}
```

OUTPUT:

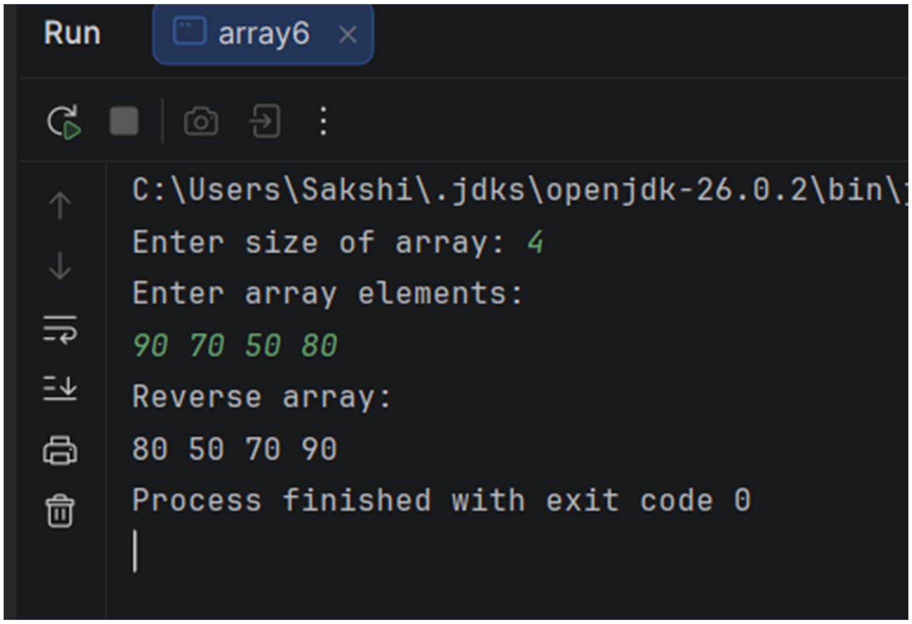


Q7.

CODE:

```
import java.util.Scanner;
public class array6 {
        public static void main(String[] args) {
            Scanner sc = new Scanner(System.in);
            System.out.print("Enter size of array: ");
            int n = sc.nextInt();
            int arr[] = new int[n];
            System.out.println("Enter array elements:");
            for(int i = 0; i < n; i++) {
                arr[i] = sc.nextInt();
            }
            System.out.println("Reverse array:");
            for(int i = n - 1; i >= 0; i--) {
                System.out.print(arr[i] + " ");
            }
            sc.close();
        }
    }
```

OUTPUT:



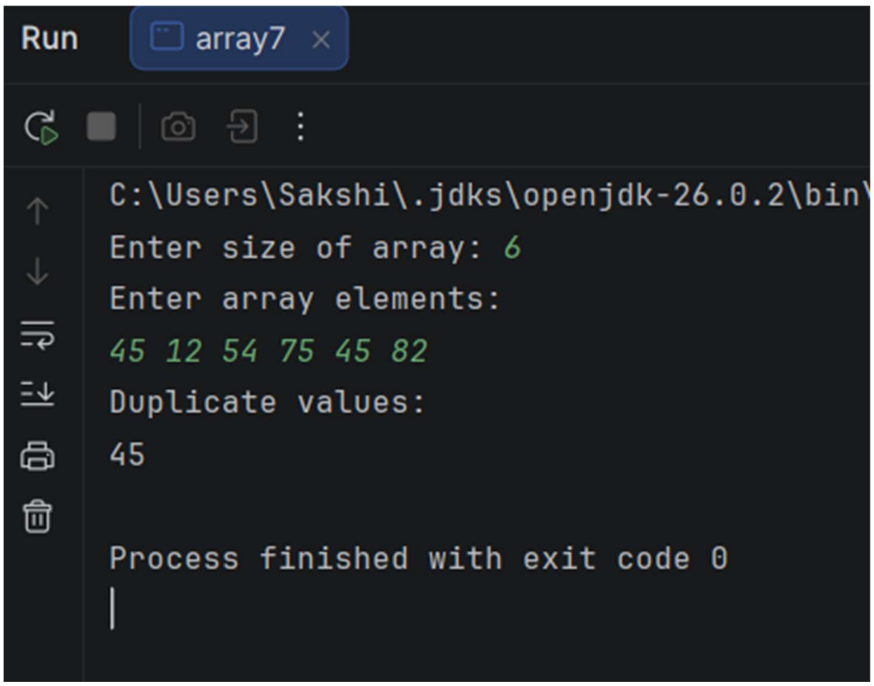
Q8.

CODE:

```
import java.util.Scanner;
public class array7 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter size of array: ");
        int n = sc.nextInt();
        int arr[] = new int[n];
        System.out.println("Enter array elements:");
        for(int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }
        System.out.println("Duplicate values:");
        for(int i = 0; i < n; i++) {
            for(int j = i + 1; j < n; j++) {
                if(arr[i] == arr[j]) {
                    System.out.println(arr[i]);
```

```
                }
            }
        }
        sc.close();
    }
}
```

OUTPUT:



Q9.

CODE:

```
import java.util.Scanner;
public class ComplexNumber {
    int number1;
    int number2;
    void setNumber1(int number1) {
        this.number1 = number1;
```

```
}
void setNumber2(int number2) {
   this.number2 = number2;
}
int getNumber1() {
   return number1;
}
int getNumber2() {
   return number2;
}
int computeComplexNumber() {
   return number1 * number2;
}
public static void main(String[] args) {
   Scanner sc = new Scanner(System.in);
   ComplexNumber arr[] = new ComplexNumber[5];
   for(int i = 0; i < 5; i++) {
      arr[i] = new ComplexNumber();
      System.out.println("Enter details for object " + (i + 1));
      System.out.print("Enter number1: ");
      int n1 = sc.nextInt();
      System.out.print("Enter number2: ");
      int n2 = sc.nextInt();
      arr[i].setNumber1(n1);
      arr[i].setNumber2(n2);
   }
   System.out.println();
   System.out.println("Results:");
   for(int i = 0; i < 5; i++) {
      int result = arr[i].computeComplexNumber();
      System.out.println("Result " + (i + 1) + " = " + result);
   }
```

```
        sc.close();
    }
}
```

OUTPUT:

```
C:\Users\Sakshi\.jdks\openjdk-26.0.2\bin\java.exe "-java
Enter details for object 1
Enter number1: 8
Enter number2: 5
Enter details for object 2
Enter number1: 9
Enter number2: 16
Enter details for object 3
Enter number1: 36
Enter number2: 84
Enter details for object 4
Enter number1: 5
Enter number2: 6
Enter details for object 5
Enter number1: 8
Enter number2: 10

Results:
Result 1 = 40
Result 2 = 144
Result 3 = 3024
Result 4 = 30
Result 5 = 80

Process finished with exit code 0
```

Q10.

CODE:

```
import java.util.Scanner;
public class array8 {
	public static void main(String[] args) {
	    Scanner sc = new Scanner(System.in);
	    System.out.print("Enter size of array: ");
	    int n = sc.nextInt();
	    int arr[] = new int[n];
	    System.out.println("Enter array elements:");
	    for(int i = 0; i < n; i++) {
	        arr[i] = sc.nextInt();
	    }
	    System.out.println("Duplicate values:");
	    for(int i = 0; i < n; i++) {
	        for(int j = i + 1; j < n; j++) {
	            if(arr[i] == arr[j]) {
	                System.out.println(arr[i]);
	            }
	        }
	    }
	    sc.close();
	}
    }
```

OUTPUT:


Run
array8
C:\Users\Sakshi\.jdks\openjdk-26.0.2\bin
Enter size of array: 5
Enter array elements:
25 48 95 75
48
Duplicate values:
48
Process finished with exit code 0